llmXive Automated Review of PerceptionDLM: Parallel Region Perception with Multimodal Diffusion Language Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the factual accuracy of claims and the validity of their supporting citations.

1. Overstated Novelty Claims In the Abstract and Introduction, the authors claim: "To the best of our knowledge, we are the first to achieve parallel region caption and perception by leveraging the advantages of diffusion language models." While the specific combination of *region-level* parallelism with *diffusion* language models may be novel, the claim of being the "first" to achieve "parallel region caption" is too broad. There are prior works exploring parallel decoding in vision-language tasks (e.g., non-autoregressive image captioning or parallel grounding) that are not sufficiently distinguished. The claim should be refined to specify "parallel region captioning using *diffusion language models*" to avoid ambiguity and potential inaccuracy regarding the broader field of parallel generation.

2. Non-Existent or Unverifiable Citations The paper relies heavily on `GPT-5.2` as a judge model for the ParaDLC-Bench evaluation (e.g., Section 4, line 138; Appendix, line 102). The citation `gpt5.2` in the bibliography points to a URL for "OpenAI-GPT-5.2" with a 2025 date. As of the current public knowledge, GPT-5.2 is not a released or publicly documented model version. Using an unreleased or non-existent model as a primary evaluation metric (the "judge") severely compromises the reproducibility and factual validity of the benchmark results. The authors must either switch to a publicly available, verifiable model (e.g., GPT-4o, Llama-3.1-70B) or provide a clear, accessible link to the specific model weights/API used.

3. Data Construction Pipeline Dependencies The training data engine section (Section 3, line 168) cites `carion2025sam` (SAM 3) for re-predicting masks. The bibliography lists this as a 2025 arXiv preprint. If this model is not yet publicly available or stable, the claim that the authors "obtained 334k images... from the COCONut dataset" using this specific tool is difficult to verify. The authors should clarify the exact version and availability of SAM 3 used in their pipeline to ensure the reproducibility of the ParaCaption-5.7M dataset construction.

4. Precision of Comparative Claims The claim in the Introduction (line 108) that "PerceptionDLM-Base outperforms LLaDA-V on 15 of 16 benchmarks" is technically supported by Table 1 (where MMMU is the exception). However, the phrasing could be slightly misleading as it implies a dominant performance across the board without immediately highlighting the specific failure case (MMMU) where the baseline is superior. While not a factual error, it is a matter of precision in scientific communication. The text should explicitly mention the MMMU exception in the same sentence to provide a balanced view of the comparative results.

5. Citation Consistency The citation `nielarge` (e.g., Section 3, line 10) refers to "Large Language Diffusion Models" (NeurIPS 2025). The bibliography entry `nielarge` is consistent, but the paper frequently cites `you2025llada` (LLaDA-V) as the primary baseline. The distinction between the base LLaDA model and the LLaDA-V variant should be clear in the text to avoid confusion about which specific architecture is being compared. The current text is generally clear, but ensuring that all citations to "LLaDA" explicitly refer to the correct variant (Base vs. V) would improve accuracy.

In summary, the paper makes strong claims about novelty and performance that are generally supported by the provided tables, but the reliance on unreleased models (GPT-5.2, SAM 3) for critical evaluation and data construction steps introduces significant risks to reproducibility and factual accuracy. The "first" claim also requires more precise wording to avoid overgeneralization.

Action items.

- [writing] The claim of being the 'first to achieve parallel region caption' (Abstract) is overstated. The authors must distinguish their work from prior non-autoregressive or parallel decoding attempts in vision-language models that may have touched on similar concepts, even if not explicitly for 'region' tasks.
- [science] The citation 'gpt5.2' (Section 4) refers to a model version that does not exist in the public domain. Citing a non-existent model as a judge for benchmark evaluation undermines the reproducibility and factual accuracy of the experimental claims. Use a publicly available model or provide a verifiable source.
- [writing] The claim that PerceptionDLM-Base 'outperforms LLaDA-V on 15 of 16 benchmarks' (Intro) is technically accurate but misleading without explicitly highlighting the specific benchmark (MMMU) where it fails, as the text implies near-universal superiority.
- [science] The citation 'carion2025sam' (Section 3) refers to 'SAM 3', a 2025 arXiv preprint. If this model is not publicly available, the claim that the authors used it to construct the dataset is unverifiable. Clarify the exact version and availability of this tool.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure fails to substantiate its caption's claim of showing 'failure cases' because the visualized masks and descriptions appear correct rather than erroneous. Additionally, the layout does not clearly associate the text descriptions with the specific highlighted regions in the image.
Action items.

- [science] Figure 1: The caption claims to show 'failure cases' of the model, but the image displays ground-truth segmentation masks (yellow bag, pink flowers) and text descriptions that appear accurate, failing to demonstrate the specific error or hallucination the figure is intended to illustrate.
- [writing] Figure 1: The text descriptions for 'Mask 1' and 'Mask 2' are not clearly linked to the visual overlays; the figure lacks arrows, labels, or a layout convention to indicate which text corresponds to which highlighted region.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and technical shorthand that are not defined at their first point of use, creating barriers for readers outside the immediate sub-field of diffusion language models.

In the Abstract, the term "AR" is used ("autoregressive generation") but the acronym itself is not defined until the Introduction. While "autoregressive" is spelled out, the subsequent use of "AR" in the Introduction (e.g., "AR decoding") assumes the reader has made the connection. It is safer to define the acronym immediately after the first full mention.

In Section 2 (Related Work) and Section 3 (Method), the acronym "RoI" (Region of Interest) is used repeatedly (e.g., "RoI-aligned feature replay," "RoI feature tokens") without ever being explicitly defined as "Region of Interest." This is a significant omission for a paper claiming to be accessible to a broader multimodal community.

In Section 3.1, the term "SFT" is used ("supervised fine-tuning (SFT)") which is good, but later in Section 3.2 and the Appendix, "SFT" appears again. More critically, Section 4.2 introduces "TPF" (Tokens Per Forward) without a clear definition of what the metric represents or how it is calculated, relying on a citation rather than an explanation.

Finally, the term "pixel unshuffle" in Section 3.1 is used as a standard operation without explanation. While standard in computer vision, it is jargon that excludes readers unfamiliar with specific image processing pipelines.

To improve accessibility, the authors should spell out all acronyms (AR, DLM, RoI, SFT, TPF) at their first occurrence in the main text and provide brief, plain-language

explanations for technical operations like "pixel unshuffle."
Action items.

- [writing] Define 'AR' (autoregressive) and 'DLM' (diffusion language model) at their first occurrence in the Abstract and Introduction. Currently, 'AR' appears in the abstract without definition, and 'DLM' is used in the intro before being explicitly spelled out as 'Diffusion Language Models' in the Related Work section.
- [writing] Replace the acronym 'RoI' (Region of Interest) with the full term 'region of interest' or 'region' on first use in Section 3 (Method) and Section 2 (Related Work). The term is used frequently without prior definition, which may confuse non-specialist readers.
- [writing] Define the metric 'TPF' (Tokens Per Forward) in Section 4.2 where it is first introduced. The text states 'we adopt the Tokens Per Forward (TPF) metric' but does not explain what the metric measures or why it is relevant to the parallel decoding claim.
- [writing] Replace the acronym 'SFT' (Supervised Fine-Tuning) with the full term 'supervised fine-tuning' in Section 3.1 (Stage 3) and Section 3.2. While common in the field, it is not defined upon first use in the main text.
- [writing] Clarify the term 'pixel unshuffle' in Section 3.1. While a standard operation in super-resolution, it is jargon that may be opaque to general readers. Consider adding a brief parenthetical explanation (e.g., 'a downsampling operation that rearranges pixels into channels').

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument for using Diffusion Language Models (DLMs) to enable parallel region captioning, contrasting this with the sequential nature of Autoregressive (AR) models. The core logical flow—identifying the sequential bottleneck in AR models, proposing DLMs as a solution due to their non-autoregressive nature, and validating this with a new benchmark—is sound.

However, there are specific logical inconsistencies and unsupported causal claims that require clarification:

1. Contradiction in Inference Mechanism: The Conclusion states the model generates captions "in a single denoising step." This directly contradicts the methodology described in Section 3 (Eq. 1) and Section 4 (Implementation Details), which explicitly define a multi-step denoising process (32 steps). A single-step generation would imply a one-shot diffusion model, which is not what is described or evaluated. This is a

significant logical error in the summary of the method.

2. Overstated Efficiency Claims: The caption for Figure 1(b) and the Introduction claim the approach "entirely avoids the linear growth in inference cost." While the *per-token* latency is constant (unlike AR), the *total* inference time still scales with the number of regions (N) because the model must process N distinct masked spans over T steps. The logic holds that it is more efficient than AR (which scales as $N \times L$), but the claim of avoiding linear growth entirely is mathematically imprecise. The latency likely scales linearly with N but with a much smaller constant factor (or sub-linearly if batched), not "avoids" it.

3. Unsupported Causal Link to Reasoning Limitations: In Section 4.1, the authors attribute the performance gap in reasoning tasks (MMMU, MathVista) to the fact that "arbitrary-order parallel decoding fundamentally limits the reasoning potential." While this is a known hypothesis in the DLM literature (cited as [ni2026flexibility]), the paper itself does not provide internal evidence or a logical derivation to support this specific causal claim for their model. The paper demonstrates *that* they are lower, but the *why* is asserted as a fundamental property of the architecture without experimental ablation or theoretical proof within the text. This weakens the logical completeness of the discussion on limitations.

4. Benchmark Logic: The ParaDLC-Bench evaluation relies on an LLM judge (GPT-5.2) to assess "inter-region feature independence." The logic assumes the judge is perfect at detecting cross-region hallucinations. While the authors test judge sensitivity (Appendix), the core logical premise that the benchmark score perfectly reflects the model's ability to disentangle regions relies entirely on the judge's capability, which is an external variable not fully controlled or proven within the paper's internal logic.

The paper is logically consistent in its primary contribution (parallelism enables speed), but the secondary claims regarding the nature of the inference steps and the fundamental limits of reasoning require more precise wording to avoid contradiction.
Action items.

- [writing] The claim that PerceptionDLM generates captions 'in a single denoising step' (Conclusion) contradicts the methodology (Eq. 1, Sec 4) which specifies a multi-step Markov process (32 steps). This logical inconsistency misrepresents the inference mechanism.

- [writing] The argument that parallel generation 'entirely avoids linear growth' (Fig 1 caption) is logically incomplete. While token-level parallelism is achieved, the total compute still scales with the number of regions (N) due to the fixed number of diffusion steps per region. The claim should be refined to 'sub-linear latency growth' or 'constant latency per region' rather than avoiding linear growth entirely.

- [science] The conclusion that 'arbitrary-order parallel decoding fundamentally limits reasoning potential' (Sec 4.1) is asserted without a causal mechanism or internal evidence in the paper. The paper demonstrates efficiency gains but does not logically derive the reasoning limitation from the parallel architecture itself, creating a gap between the premise (parallelism) and the conclusion (reasoning failure).

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims that slightly overreach the direct evidence provided, primarily regarding the uniqueness of the contribution and the interpretation of performance gains.

First, the assertion in the Abstract and Introduction that the authors are the “first to achieve parallel region caption and perception by leveraging the advantages of diffusion language models” is a strong claim. While the specific application of DLMs to this task is novel, the concept of parallel generation for multiple regions is not entirely new in the broader context of non-autoregressive models or parallel decoding strategies in other domains. The claim should be tempered to explicitly state “first to achieve... using diffusion language models” to avoid implying a broader novelty that might not be fully supported.

Second, the claim that PerceptionDLM “nearly doubles the performance” of LLaDA-V (62.4% vs 35.2%) on ParaDLC-Bench (Introduction, Section 4.2) requires nuance. The comparison is between a model designed for parallel generation (PerceptionDLM, TPF=2.9) and a baseline evaluated sequentially (LLaDA-V, TPF=1). The significant performance gap may be partially attributed to the architectural mismatch in how the models handle the multi-region task (parallel vs. sequential) rather than a pure doubling of the underlying model’s capability. The text should clarify that the comparison highlights the benefit of the parallel paradigm in this specific setting, rather than implying a universal doubling of model quality.

Finally, the statement in Section 4.2 that “arbitrary-order parallel decoding fundamentally limits the reasoning potential of diffusion language models” is a strong theoretical claim. While the paper cites external work (ni2026flexibility), the authors should ensure they are not overgeneralizing this limitation to their specific architecture or all reasoning tasks without providing their own analysis or ablation that directly supports this conclusion in the context of their model. The current text presents it as a definitive fact derived from their work, which may be an overreach.

Action items.

- [writing] The claim of being the ‘first to achieve parallel region caption’ (Abstract, Intro) is overreaching. Qualify this to ‘first to achieve... using diffusion language models’ to avoid implying broader novelty not supported by the data.

- [writing] The claim that PerceptionDLM 'nearly doubles the performance' of LLaDA-V (62.4% vs 35.2%) on ParaDLC-Bench (Intro, Exp) conflates architectural parallelism with model capability. Clarify that the gap reflects the parallel vs. sequential paradigm difference, not just inherent model quality.
- [writing] The statement that 'arbitrary-order parallel decoding fundamentally limits reasoning potential' (Exp, Sec 4.2) is a strong theoretical claim. Ensure this is not over-generalized from the cited work without specific ablation or analysis supporting this limitation in the authors' own architecture.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a novel approach to parallel region perception using Diffusion Language Models (DLMs). From a safety and ethics perspective, the work is generally sound as it focuses on technical efficiency and does not explicitly target high-risk dual-use applications (e.g., surveillance, deepfakes, or autonomous weapon targeting). However, several critical omissions in the current draft require attention before publication.

First, the Ethics Statement is currently commented out in `main.tex` (lines 68-75). While the authors note in the comments that they do not collect new data on humans, the model is trained on and evaluated against datasets containing human and animal imagery (e.g., Objects365, COCONut, SA-1B). The generated outputs are detailed textual descriptions of these regions. An active, formal ethics statement is mandatory to explicitly address: (1) the potential for the model to inadvertently reveal sensitive attributes (e.g., identity, location, health status) if applied to real-world images; (2) the handling of bias in the training data; and (3) the scope of the generated content to ensure it does not facilitate harassment or doxxing. The current placeholder is insufficient for a paper involving multimodal perception of potentially sensitive visual content.

Second, the construction of the ParaDLC-Bench relies heavily on an LLM judge (GPT-5.2) for generating evaluation questions and attributes. While the authors mention human verification, the specific prompts used to guide the LLM (referenced as `figs/Prompt_*.pdf` in the appendix) are not fully transparent in the text. To ensure reproducibility and mitigate the risk of "judge bias" (where the evaluation metric inherits the biases of the judge model), the authors should either include the full prompt text in the main body or provide a direct, stable link to the prompt repository. Furthermore, the human verification process needs more detail: what specific criteria did human annotators use to filter out biased or harmful questions generated by the LLM?

Third, the data provenance for the training set (ParaCaption-5.7M) requires clarification regarding consent and licensing. The pipeline uses GAR-8B to generate captions for masks from SA-1B and COCONut. While these are open-source datasets, the creation of a new, large-scale derivative dataset (5.7M captioned masks) raises questions about the compatibility of the original licenses with the new dataset's intended use (e.g., commercial vs. research). The authors should explicitly state the licenses of the source datasets and confirm that the automated captioning process and the resulting dataset comply with these licenses.

Finally, while the paper discusses "hallucination" as a technical failure mode, it should briefly address the safety implications of hallucination in this specific context. If the model hallucinates attributes for a region containing a person (e.g., attributing a weapon to a harmless object), this could have downstream safety consequences if the model is deployed in security or monitoring applications. A brief discussion on mitigating these risks in future deployments would strengthen the ethical robustness of the paper.

In summary, the technical contribution is significant, but the manuscript must include an active ethics statement, clarify the LLM judge prompts and verification process, and explicitly address data licensing and the safety implications of hallucination in region captioning.

Action items.

- [writing] The Ethics Statement section is commented out in main.tex (lines 68-75). Given the use of human-centric datasets (Objects365, COCONut) and the generation of detailed descriptions for human/animal regions, an active, explicit ethics statement is required to address potential biases, privacy implications, and the scope of generated content.
- [writing] The benchmark construction (ParaDLC-Bench) relies on GPT-5.2 for attribute extraction and question generation. The manuscript must explicitly disclose the specific prompts used (currently only referenced as figures in the appendix) and detail the human-in-the-loop verification process to ensure the evaluation questions do not encode model biases or hallucinations.
- [writing] The training data pipeline (ParaCaption-5.7M) utilizes GAR-8B to generate captions for SA-1B and COCONut masks. The authors must clarify the consent status of the source images in these datasets and confirm that the automated generation of detailed descriptions does not violate the original data licenses regarding derivative works or commercial use.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling architectural shift for region captioning, leveraging the parallel nature of Diffusion Language Models (DLMs) to overcome the sequential bottlenecks of autoregressive (AR) baselines. The central claim—that PerceptionDLM achieves competitive accuracy with significantly improved throughput in multi-region settings—is supported by a substantial body of experimental data, including a new benchmark (ParaDLC-Bench) and extensive ablation studies.

However, the scientific evidence regarding the efficiency claims requires stronger statistical grounding. The reported 3.44x throughput speedup and the specific TPF (Tokens Per Forward) metric of 2.9 are derived from single-run or limited-run latency measurements (Section 4, "Evaluation on Captioning Benchmarks"). In high-performance computing contexts, inference latency can fluctuate due to GPU thermal throttling, memory bandwidth contention, or kernel launch overhead. The manuscript currently presents these efficiency gains as deterministic facts without reporting standard deviations, confidence intervals, or the number of independent trials conducted. To robustly support the claim of "substantial speed improvements," the authors should provide a statistical analysis (e.g., mean ± std over 10+ runs) and a significance test (e.g., paired t-test) comparing PerceptionDLM against the AR baselines (GAR, DAM) under identical hardware conditions.

Furthermore, the validity of the evaluation metric relies heavily on the LLM-as-a-Judge paradigm. While the authors demonstrate robustness across different judge models (GPT-5.2, Qwen3.5, Gemini) in the Appendix, the primary results are entirely dependent on the GPT-5.2 judge. The "Negative & Interference" questions, which are critical for the paper's claim of reduced hallucination, are subjective and complex. Without a human evaluation subset to establish the correlation between the LLM judge's scores and human judgment, there is a risk that the reported accuracy gains (e.g., 62.4% vs. 35.2% for LLaDA-V) reflect biases in the judge model rather than true model performance. A small-scale human evaluation (e.g., 50-100 samples) is necessary to validate the benchmark's reliability.

Finally, the ablation study regarding "Region Prompting" (Appendix, Table 4) shows a catastrophic performance drop to 1.1% accuracy when this component is removed. While this highlights the component's importance, such an extreme sensitivity warrants a deeper investigation into potential confounding factors. Is the failure due to a lack of spatial grounding, or does the model architecture collapse without this specific embedding injection? A qualitative error analysis of the "w/o region prompting" outputs would strengthen the evidence that the failure is strictly due to the missing mechanism and not an implementation artifact or training instability.

Overall, the evidence is strong but requires statistical rigor in efficiency reporting and validation of the evaluation protocol to fully support the central claims.

Action items.

- [science] The efficiency claims (3.44x speedup) rely on TPF=2.9 but lack statistical validation. Report mean/std latency over multiple runs and a significance test (e.g., t-test) against the AR baseline to rule out variance or hardware noise.
- [science] The ParaDLC-Bench evaluation relies entirely on GPT-5.2 as a judge. While Appendix A shows robustness to Qwen3.5, the primary results lack human verification. Include a human evaluation subset (e.g., 50-100 samples) to validate the LLM judge's

correlation with ground truth.

- [science] The ablation study in Table 4 (Appendix) shows a catastrophic drop to 1.1% accuracy without region prompting. This extreme sensitivity suggests a potential confounding variable or implementation artifact. Provide a qualitative analysis or error breakdown to confirm the failure mode is strictly due to missing prompts.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis supporting the claims of efficiency and performance in this paper requires significant strengthening to meet publication standards.

First, the core claim of "substantial speed improvements" (Abstract; Section 4.2) relies on single-point latency measurements presented in Table 2 and Figure 1(b). The paper reports a specific inference time (e.g., 276s vs 479s) and a throughput speedup (3.44x) without any measure of variance. Diffusion models involve stochastic sampling processes, and hardware inference times can fluctuate. Without reporting the mean, standard deviation, and sample size (N) from multiple independent runs (e.g., N=5 or N=10), it is impossible to determine if the observed speedup is a robust effect or a result of random variance. The authors must re-run the efficiency benchmarks with multiple seeds and report confidence intervals.

Second, the primary evaluation metric for the ParaDLC-Bench is an "Average Accuracy" derived from an LLM judge (GPT-5.2). While the authors acknowledge judge sensitivity in the Appendix (Table 4), the main results (Table 2) present the GPT-5.2 scores as definitive. There is no statistical test (e.g., bootstrap resampling or permutation tests) to confirm that the difference between PerceptionDLM (62.4%) and the best baseline (GAR, 69.5% on Pos, but lower on Avg) is statistically significant. Given the potential for LLM judges to exhibit bias or high variance on specific prompts, the authors should provide confidence intervals for the reported accuracy scores.

Finally, the ablation studies in the Appendix (Tables 3-6) present performance differences as absolute percentage points (e.g., "degrades by 6.2%"). Without p-values or effect sizes, these claims lack statistical grounding. The authors should apply appropriate statistical tests (e.g., paired t-tests or Wilcoxon signed-rank tests) to the ablation results to validate that the architectural components (Region Prompting, Structured Attention) contribute significantly to performance rather than observing noise.
Action items.

- [science] The efficiency comparison in Table 2 and Figure 1(b) lacks statistical rigor. The reported 'Time (s)' and 'TPS' are single-point measurements without confidence intervals, standard deviations, or sample sizes (N). Given the stochastic nature of diffusion sampling and hardware variance, a single run is insufficient to claim a '3.44x speedup'. Please report mean and standard deviation over at least 5 independent runs per configuration.

- [science] The evaluation metric 'Avg (%)' in Table 2 is derived from an LLM judge (GPT-5.2) without reporting inter-rater reliability or sensitivity analysis. While Appendix Table 4 shows results with different judges, the main text treats the GPT-5.2 score as ground truth. Please include a statistical test (e.g., bootstrap confidence intervals) to demonstrate that the observed performance gap (62.4% vs 35.2%) is statistically significant and not an artifact of the specific judge model's bias.
- [science] In the ablation studies (Appendix Tables 3-6), the authors report percentage point differences (e.g., '6.2% drop') but do not provide p-values or effect sizes to determine if these differences are statistically significant. With the large dataset sizes implied (millions of samples), even trivial differences may appear significant, or conversely, small but meaningful improvements may be dismissed. Please add statistical significance testing for all ablation comparisons.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical sophistication, and the writing is generally clear and professional. The logical flow from the introduction of the problem to the proposed solution and evaluation is well-structured. However, there are a few minor issues regarding grammar, typos, and sentence structure that, while not critical, detract slightly from the overall polish of the paper.

Specifically, in the Abstract, the phrase "parallel region caption and perception" lacks the gerund form for "caption," breaking the parallel structure with "perception." Changing this to "captioning and perception" would improve readability. In Section 4, the caption for Table 1 contains a clear typo: "officiall checkpoints" should be "official checkpoints." Additionally, in the Appendix, the reference to the training parameters table uses the label `traing_params` (missing the 'i'), which should be corrected to `training_params` to match the likely intended label and ensure consistency.

There are also minor opportunities to improve sentence flow. For instance, in the Introduction, the transition between the limitations of autoregressive models and the potential of diffusion models could be slightly smoother. The sentence "Yet directly extending diffusion-based VLMs to fine-grained localized perception remains non-trivial" is strong, but the preceding sentence could be tightened to avoid a slight redundancy in explaining the sequential nature of AR models.

Overall, the writing quality is strong, but addressing these specific typos and minor grammatical inconsistencies will enhance the professional presentation of the work.

Action items.

- [writing] In Section 3 (Method), the phrase 'relying on autoregressive (AR) decoding' is used, but the sentence structure in the paragraph beginning 'Despite high accuracy...' is slightly repetitive. Consider varying the sentence structure to improve flow.
- [writing] In the Abstract, the sentence 'To the best of our knowledge, we are the first to achieve parallel region caption and perception...' is grammatically slightly awkward. Consider rephrasing to '...parallel region captioning and perception...' for better parallelism.
- [writing] In Section 4 (Experiment), the phrase 'official1 checkpoints' appears in the caption of Table 1. This is likely a typo (extra '1') and should be corrected to 'official checkpoints'.
- [writing] In the Appendix, the phrase 'The four-stage training setup of PerceptionDLM-Base is listed in~\Cref{tab:traing_params}' contains a typo in the table label name ('traing_params' vs 'training_params'). Ensure consistency with the actual label defined in the LaTeX source.